



A HARDWARE-CENTERED SDK

NEO·GEO SDK

Direct 68000-side control over **VRAM, palettes, sprites, DMA, FIX tiles**, and a custom YM2610 sound stack driven by a Z80 sound driver.

68000 · Z80 · YM2610

INSERT COIN

THE PITCH

Real Neo Geo development, made practical.

For self-made **games, demos, and homebrew** that want to stay close to original Neo Geo development practice, while still using modern build tooling, scripting, and emulator-based test loops. The goal is not to abstract the hardware away. The goal is to make working with it possible on Linux, WSL, and Windows.

Target	SNK Neo Geo, cartridge-era arcade hardware (MVS) and the home AES.
Release	v1.7.0 · MIT licensed.
Host platforms	Linux · WSL · Windows.
Build languages	C99 · C++14 · 68000 asm · Z80 asm · Python.
Audio layers	FM · SSG · ADPCM-A · ADPCM-B.
Shipped games	6, all building to real ROMs.
Engine builds	2, plain C and C++14, same ABI.

THE TARGET MACHINE

Three chips. One bus. **No engine.**

MAIN CPU

M68000 @ 12 MHz

Game logic, VRAM, palettes, sprites, DMA, FIX tiles, directly.

SOUND CPU

Z80A @ 4 MHz

Runs the dedicated driver; talks to the YM2610 over command ports.

SOUND CHIP

YM2610

FM synthesis, SSG square waves, ADPCM-A and ADPCM-B sample playback.

THE DEVELOPMENT MODEL

Graphics, palettes, FIX tiles, ROM layout, and sound commands are part of the day-to-day programming model rather than hidden behind a large engine layer. The SDK preserves that: it gives you headers, build glue, asset pipelines, and a 2D engine you can opt into, not abstractions you cannot see past.

01

For everyone.

NEO GEO DEV 101

Before the SDK matters, the hardware has to make sense. Six sections on what every Neo Geo programmer needs to know: the FIX layer, sprites, palettes, VRAM access, interrupts, and how the two CPUs talk.

HOW A FRAME IS DRAWN

A frame is **stacked**, not composited.

P1

FIX layer

40 × 32 grid of 8×8 tiles in a fixed location, 28 rows visible. Up to 15 colours per cell, 16 palettes, 4096 tiles. This is the HUD / text plane, and it is always on top.

128 KB ROM
S1

P2

Sprites

Up to 380 on screen, 96 per scanline. Each sprite is a vertical strip of up to 32 tiles of 16×16 pixels, with per-sprite reduction and a chain bit that welds a strip to its left neighbour.

UP TO 128 MB ROM
C ROMS

P3

Backdrop

A single solid colour behind everything else. One value from 65,536. That is the entire background layer.

1 COLOUR

RULE ONE

SCB2 can only shrink, never stretch

Author art at its largest intended size.

RULE TWO

HIGHER slot number draws in front

Backgrounds belong at slots 1 to 32, characters at 96 to 223, foreground effects at 288 and up.

VRAM ACCESS VIA LSPC

You don't poke VRAM directly.

ADDRESS	R/W	REGISTER
\$3C0000	W	VRAM Address (16-bit word index)
\$3C0000	R	Read Data (address register unchanged)
\$3C0002	W	VRAM Write Data. Writing pushes to VRAM
\$3C0004	R/W	Auto-Increment. Added to the address after every write
\$3C0006	W	Mode register. Timer / IRQ / animation speed
\$3C0008	W	Timer High
\$3C000A	W	Timer Low
\$3C000C	W	Interrupt Clear (bit 1: timer, bit 2: VBL)

VRAM SIZE

64 KB + 4 KB

All access is word-aligned. Long-word and byte access are not allowed.

TIMING RULE

~1.4 μs

Wait at least 16 CPU clocks after touching the Address register before reading.
Cross-bank writes cannot use auto-increment.

SDK SHORTCUT

Headers in sdk/

The SDK wraps these registers in C macros so you write to named symbols, not raw addresses.

TWO INTERRUPTS. THAT'S THE WHOLE FRAME LOOP.

Where your code **runs**.

IRQ 1 · V-BLANK

Vertical blanking

Fires once per frame, vector \$64. This is your main game tick: read inputs, run logic, push the frame's sprite list into VRAM. The window is about 3 ms.

```
MOVE.W #4, $3C000C ; ack VBL
MOVE.B D0, $300001 ; kick watchdog
```

IRQ 2 · TIMER

Raster timer

Fires when the LSPC timer counter hits 0, vector \$68. Decrements every 167 ns (one pixel). Used for raster effects: split scrolls, mid-frame palette swaps, status bars.

```
MOVE.W #2, $3C000C ; ack timer
```

TIP

For "every N pixels," set Mode register bits 4 through 7 to %1001 and Timer = N minus 1. For multiple arbitrary scanlines, use mode %1101 and reload Timer inside the handler.

ONE BYTE AT A TIME

How the two CPUs talk.

68000 → Z80

\$320000 (write byte)

The game writes a 1-byte sound code. The Z80 reads it on its own port and dispatches to the right music / SFX / FM / SSG handler.

Z80 → 68000

\$320000 (read byte)

The Z80 can post a status byte back; the game reads from the same address.

SDK ABSTRACTION

You don't write that byte by hand. You call **playMusic()** / **playSFX()** with a named id from `sdk/sound_ids.h`. The SDK encodes it; the Z80 driver decodes it.

ONE CAVEAT

A prefix byte and its parameter are two separate NMIs on the Z80. Space multi-byte commands with a `waitVbl()` or the second one can be dropped.

COLOUR IS PALETTED, NOT DIRECT

4,096 on screen, of 65,536.

15 colours per character, 4 bits of palette index, colour 0 is transparent.

256 palettes total: 16 reserved for FIX, 239 for sprites, 1 for the backdrop.

4,096 colours on screen, of a 16-bit pool of 65,536.

Two palette banks, toggled via \$3A000F / \$3A001F for snappy palette swaps.

Shadow bit at \$3A0011 dims the entire display: pause screens, hit flashes.

The backdrop register is the last word of palette RAM, \$401FFE. \$402000 is a mirror that silently does nothing.

PALETTE RAM

\$400000–\$401FFF

Two banks, accessed as words/longs. The SDK exposes them as arrays in the C runtime.

THE COLOUR WORD IS NOT RGB444

5 bits per channel, 32 levels, steps of 8

Bit 15 is the dark bit, bits 14 to 12 carry the LSB of each channel, bits 11 to 0 carry the top four bits of R, G and B. The artbox quantiser snaps to exactly this lattice.

DESIGN INTENT

The machine is close to the metal by design.

The SDK doesn't try to change that.

PRESERVE

Original Neo Geo development practice: direct hardware register access.

MODERNISE

Build tooling, asset scripting, and emulator-based test loops on current operating systems.

RESIST

Heavy abstraction layers that hide what the hardware is actually doing.

REPOSITORY MAP

The repo, end to end.

```
neogeosdk/
├─ sdk/                68000 headers, runtime, both 2D engines
│  └─ 2d_engine/        the C engine      (ng_*.c / ng_*.h)
│  └─ 2d_engine_plus/   the C++ engine    (ng_*.cpp / ng_*.hpp), same ABI
│  └─ ng_fix/ ng_audio/ ng_video/ ng_scene/ ng_show/ bsp/
│  └─ neogeo.h macro.h  hardware registers and inline helpers
│  └─ sound_ids.h       named music / SFX / FM / SSG ids
├─ games/              one directory per game, self-contained
│  └─ demo/              the 25-chapter engine reel      (id 777)
│  └─ demo_plus/         C++-engine smoke test          (id 778)
│  └─ skylance/          Sky Lance, a complete shooter  (id 779)
│  └─ helloworld/        minimal FIX text + one sample  (id 772)
│  └─ tutorial/          minimal engine loop            (id 555)
│  └─ neogeogame/        empty starting point           (id 775)
├─ artbox/              PNG → sprite / FIX / C-ROM pipeline + Artbox Studio
├─ sound/               YM2610 stack: Z80 driver, MML, FM, SSG, samples
├─ docs/                the documentation set
├─ tools/               validators, the API generator, the manual builder
├─ dist/ roms/ hash_eagle/ build output and MAME hash sets
├─ Makefile              Linux + WSL build
└─ MakefileWin32.mak     native Windows build
```

TOP-LEVEL ENTRY

main.c • user.c

The 68000-side game flow and the BIOS-hook startup handlers. main.c is the file you edit; user.c you almost never touch.

ASSET PIPELINES

artbox/ + sound/

Two parallel pipelines: PNG to sprite C-ROMs and FIX, plus WAV and MML to ADPCM and driver tables.

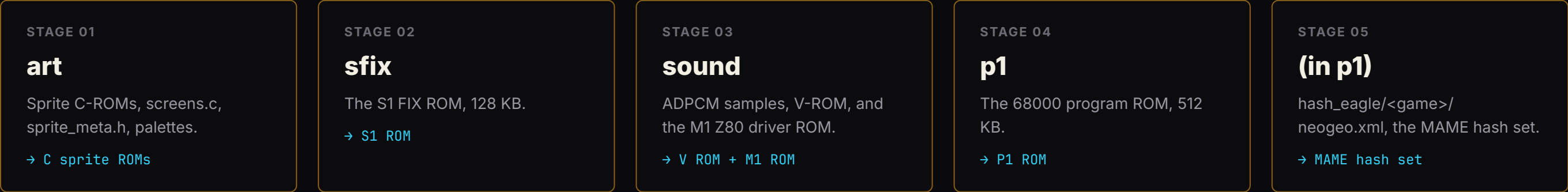
OUTPUT TARGET

roms/<game>/

Generated ROM sets kept in-tree so MAME test runs are instant.

END-TO-END PIPELINE

Five stages. One **make all**.



```
make GAME=<name> GAME_CFG_FILE=games/<name>/game.cfg all
make GAME=<name> GAME_CFG_FILE=games/<name>/game.cfg test
```

THE GAME.CFG CROSS-CHECK

Every game carries its own game.cfg, and the build cross-checks GAME= against it, which is what stops you silently building one game’s art into another game’s ROM. In day-to-day work you run p1 almost exclusively; it takes seconds.

THREE HOST PLATFORMS, ONE SDK

Cross-platform **without compromise.**

LINUX

PRIMARY

x-tools-v2/

m68k-unknown-elf cross compiler bundle. Legacy x-tools/ still picked up as a fallback if v2 is absent.

REQUIRES

- git, mame, srecord
- cmake, build-essential
- python3 + numpy / Pillow / pypng
- wla-z80, wlalink
- PyQt6 (Artbox / Sound Studio)

WSL

USE LINUX

Makefile

Use the Linux Makefile, not the Win32 one. A documented WSL2 + Ubuntu 24.04 + PulseAudio path covers audio playback.

REQUIRES

- Linux x-tools-v2/m68k-unknown-elf
- Linux python3, wla-z80 / wlalink
- /mnt/c/... checkouts work

WINDOWS

WIN32 .MAK

m68k-elf

Native build via MakefileWin32.mak, plus one-shot installers for four host setups.

REQUIRES

- m68k-elf GCC toolchain
- GNU Make on PATH
- wla-z80, wlalink, mame

C++ BUILD FLAGS

-std=c++14 -fno-exceptions -fno-rtti -fno-threadsafe-statics. No standard library, no malloc, no floating point, no order-dependent global constructors.

WHAT YOU CAN BUILD

Every entrypoint, one table.

TARGET	RESULT
all	art + sfix + sound + p1
p1	68000 program ROM only
aes / mvs	Force the AES or MVS platform variant
art / art-crt	Rebuild sprite C-ROMs, or via the CRT pipeline
sfix / srom	Rebuild the S1 FIX ROM
sound	samples + vrom + fmpatches + fm + mml + ssgconfig + ssg + m1rom
m1rom / m1rom-c	The shipping ASM driver, or the experimental C driver

TARGET	RESULT
compare-driver	Build both M1 drivers and diff the ROM images
dist	Package dist/roms/<game>.zip
test / test-aes	Run in MAME (MVS sp-s2.sp1, or AES unibios22)
debug / debug-build	MAME with the debugger, or P1 with symbols + map
gdb / gdb-trace / gdb-remote	GDB against the running ROM
bios-list / games-list / menu / help	Discovery targets
clean / sound-clean / art-clean / clean-all	Cleanup targets

02

For the programmer.

THE ENGINE, THE PIPELINES, THE BUDGET

164 commits on the neo_universal_2d branch: a complete 2D engine in C and C++14, a rebuilt YM2610 driver, a rebuilt art pipeline, and the multi-game workflow that ties six shipped games together. The largest release this SDK has had.

THE GAME CONTRACT

What you actually write.

A Neo Geo cartridge boots into the BIOS, not into main(). The BIOS calls back for power-on, eye-catcher, game, and title; user.c provides those hooks. What you write is two functions.

```
void game_boot(void);    /* once, after the BIOS hands over */  
void game_frame(void);   /* once per frame, forever */
```

THE MOST COMMON FIRST BUG

game_frame() returns every frame, so anything that must persist is static or file-scope. A plain local is gone before the next call.

SAME PUBLIC ABI, TWO LANGUAGES

The same engine, shipped twice.

sdk/2d_engine/ in plain C, sdk/2d_engine_plus/ in C++14. Same design, same public extern "C" ABI, so a C game can link the C++ build and vice versa. One engine per ROM, selected with USE_2D_PLUS=1.

Frame tick	ng_game_interrupt.h	The whole engine tick, with five hook points
Characters	ng_chars.h	64-object pool: movement, animation, hitboxes, damage
Actions	ng_actions.h	Table-driven action scripts with sound and FX hooks
NPCs	ng_npcs.h	Patrol bounds, home position, think callbacks
Physics	ng_physics.h	Gravity, drag, solid rectangles, grounded tests
Level / Camera	ng_level.h / ng_camera.h	World bounds, scroll, follow, dead zone, shake
Joystick	ng_joystick.h	Edge detection, repeat, motion specials

SAME PUBLIC ABI, TWO LANGUAGES

Everything that reaches **the screen.**

Render queue	<code>ng_render_queue.h</code>	128-slot VBlank-safe deferred VRAM / palette writes
Sprite groups / pool	<code>ng_sprite_group.h</code> / <code>ng_sprite_pool.h</code>	Dirty-flag chains and the hardware slot map
FIX / Backgrounds	<code>ng_fix.h</code> / <code>ng_bg.h</code>	Text layer with a dirty-cell cache; two parallax layers
Palette FX / Particles	<code>ng_palette_fx.h</code> / <code>ng_particles.h</code>	Fade / flash / pulse / cycle; a 32-slot pool
Feedback / Depth FX	<code>ng_feedback.h</code> / <code>ng_depthfx.h</code>	Hitstop, shake, flash in one call; perspective and fog
Fixed point	<code>ng_fixed.h</code>	16.16 arithmetic, sin/cos and shrink tables

FULL API

Every module here, plus the C++14 class wrappers, is covered call by call in the 2D Engine API Reference.

PNGS IN, NEO GEO ROMS OUT

The Lab-space quantiser.

Drop PNGs into games//artbox/in// and run make art. Four ideas moved mean colour error from dE 9.10 to 8.33 and p95 from 18.19 to 15.66, with measured dither speckle down a third.

MATCH IN LAB, NOT RGB

Palette selection and remap run in CIE-Lab

Through a precomputed 32-cubed lattice LUT. Matching in one space and measuring in another is what makes a quantiser that scores well look wrong.

REFIT ONTO THE HARDWARE GRID

Lloyd iterations with the snap inside the loop

k-means picks colours in continuous space; snapping to the 32-level lattice afterward moves every centroid. Keep a move only when error actually drops.

BLUE-NOISE DITHER, SELECTIVELY

A dead band skips pixels within 18% of an endpoint

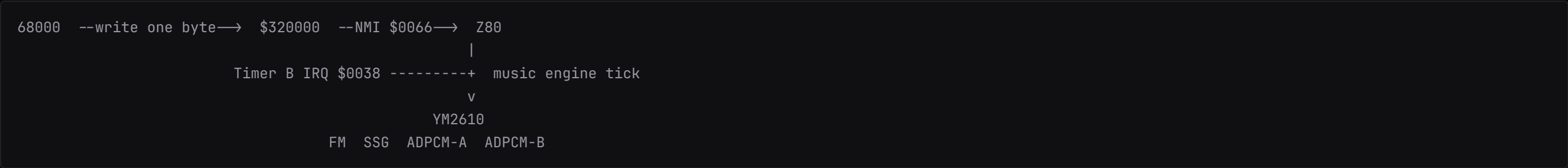
Those are the pixels where dithering adds speckle without adding colour.

ASSET ORDERING

Categories are processed as backgrounds, characters, effects, eyecatcher, npc, screens, titles, then any other directory alphabetically. An asset id is its 1-based position in that sequence, so inserting a file mid-category rennumbers everything after it.

EVERYTHING THAT MAKES NOISE RUNS ON A SECOND CPU

Audio: the stack.



ADPCM-A

6, one-shot

Short sound effects, voice samples.

ADPCM-B

1 stream

Backing tracks, ambience, long cues.

SSG

3 square waves

MML melody, plus a standalone sequencer.

FM

16-patch bank

Music, stings, formant effects, compiled sequencer.

WHAT YOU CALL FROM MAIN.C

The sound driver.

```
soundInit();                /* once at boot */
playMusic(SOUND_MUSIC_E);   playSFX(SOUND_SFX_7);
playSFXB(track);            playFMTrack(SOUND_FM_A);
playSSGTrack(SOUND_SSG_B);  soundSetSSGPreset(3);
soundFadeOut();             soundSceneReset();
speakText("READY!");
```

CHIP-LEVEL CONTROL

ADPCM-B pan, FM LFO, SSG noise

Register \$11, \$22, \$06, plus CSM (Composite Sine Mode), which auto-keys FM channel 2 from Timer A for formant and vowel-slide effects.

PRACTICES THAT KEEP AUDIO STABLE

Reset on scene change, space multi-byte commands

Use soundSceneReset() rather than stacking stops; fade before you stop; budget ADPCM-A the way you budget sprites, six channels, and a seventh trigger steals one.

STAYING INSIDE THE FRAME

16.7 ms. That's the budget.

RESOURCE	LIMIT
CPU	~12 MHz 68000, ~200,000 cycles per frame
VBlank	~3 ms, all VRAM writes must fit
Sprites	380 total, 96 per scanline
Palettes	16 entries per bank, 256 banks
FIX	40 × 32 cells, 28 rows visible, 16 palettes

No malloc during gameplay. Every engine subsystem uses a fixed pool: characters 64, NPCs 32, particles 32, palette FX 8, render queue 128.

No float or double. There is no FPU; every float costs hundreds of cycles in software emulation. Use 16.16 fixed-point.

No division in the frame loop. DIVS/DIVU cost 76 to 158 cycles each. Use the sin/cos/shrink tables and right shifts.

Write only what changed, and queue every VRAM write. Sprite groups carry dirty flags; post to the render queue and let ng_render_queue_flush() drain it in VBlank.

SIX GAMES, ONE BUILD SYSTEM

The shipped games.

ID	GAME	ENGINE	WHAT IT IS
777	demo	C	The 25-chapter engine reel
778	demo_plus	C++	The same engine through the C++ API
779	skylance	C	Sky Lance, a complete vertical shooter
772	helloworld	none	FIX text and one sample; the tutorial target
555	tutorial	C	The minimal engine loop
775	neogeogame	C	An empty starting point

SKY LANCE

Three pilots, ROOK (Blue Lance), KIRA (Green Wing), BLAZE (Red Striker), differentiated by fire rate, speed, and single vs. twin bolt. Seven stages, each running five to seven squadrons before its boss: Crimson Keep, Iron Tide, Sol Core, Night Razor, Rotor Nest, Earth Hammer, Spire God. The playfield is the 256px column the backdrop covers, not the full 320px screen, because a page is 16×16 tiles, and its backdrops had to tile seamlessly since the sprite chip can shrink but never stretch.

EVERY SUBSYSTEM, IN ORDER

25 chapters, one reel.

Each chapter isolates one subsystem, with the chapter number printed top-right so a problem can be reported by number.

ADVANCE

A

Moves to the next chapter.

RESTART

C

Restarts the current chapter.

BOOT · TITLE · FIX LAYER · FIX FX · SPRITE SCREENS · CHARACTERS · CHAR SELECT · PHYSICS · CAMERA LAB · PALETTE FX · PARTICLES · PARTICLE LOAD · FEEDBACK · DEPTH FX · DEPTH PARALLAX · NPCS ·
MINI-GAME · JOYSTICK · SCROLL LEVEL · CHAR 2D · TARGET RANGE · DEPTH RIDE · SOUND · SKY LANCE · CREDITS

WHERE THINGS GO ON DISK

One env var. **\$SDKHOME**.

```
# 1. install the toolchain
install/install-linux.sh      # or install-ubuntu.sh / install-wsl.bat
                              # or install-windows.bat

# 2. check it
m68k-unknown-elf-gcc --version
wla-z80 --version
python3 --version             # 3.10+, with Pillow and NumPy

# 3. build and run
make GAME=helloworld GAME_CFG_FILE=games/helloworld/game.cfg all
make GAME=helloworld GAME_CFG_FILE=games/helloworld/game.cfg test
```

DEFAULT BEHAVIOUR

Set SDKHOME to the checkout. See docs/DEPENDENCIES.md and the installation sections of README.md, including the WSL2 + Ubuntu 24.04 + PulseAudio walkthrough for working audio under WSL.

READ THIS BEFORE YOU FILE A BUG

Common symptoms.

SYMPTOM	CAUSE
Sprite invisible behind the background	Slot priority, HIGHER is in front
Sprite too large, can't shrink further	SCB2 only reduces; re-author the art
Backdrop colour ignored	Wrote \$402000 instead of \$401FFE
Text draws as solid blocks	The BIOS space glyph is opaque; use ng_fix_blank_cell()
Pool fills and spawns stop	Objects marked destroyed but never ng_chars_remove()d
Second byte of a sound command lost	Prefix and parameter not spaced by waitVbl()
Sprites dropped along one row	The 96-per-scanline limit
Green self-test screen on boot	The BIOS self-test, not your ROM; try BIOS=unibios40
Tearing or corrupt tiles	VRAM written outside VBlank instead of queued

THIS DECK IS THE OVERVIEW

The deep material lives in its own manuals.

INTRODUCTION.md

What the SDK is, and the reading order.

HELLO_WORLD.md

Your first ROM, in ten minutes.

PROGRAMMERS_MANUAL.md

Build, layout, engine tick, assets, budget.

API_2D_ENGINE_C.md / CPP.md

Every call, both engine builds.

SDK_API_GUIDE.md

The bare-metal 68000 helpers.

SOUND_DRIVER.md

The Z80 driver, the YM2610, the pipelines.

ARTBOX_PIPELINE.md

PNG to C-ROM, the quantiser, Artbox Studio.

GAMES.md / DEMO_CHAPTERS.md

What each game demonstrates.

performance_rules.md / GDB_GUIDE.md

Staying in budget; debugging on real timing.

164 COMMITS ON NEO_UNIVERSAL_2D

What v1.7.0 adds.

AREA	WHAT ARRIVED
2D engine	35 modules, built in thirteen staged passes: characters, actions, physics, NPCs, camera, level, particles, feedback, depth FX, palette FX, sprite groups, render queue, FIX, input, timers, events, fixed-point math.
C++14 engine	The same engine with member methods and singletons, identical extern "C" ABI, selected with USE_2D_PLUS=1.
Multi-game builds	Per-game id, ROM folder, artbox, sound tree, linker script, and MAME hash set, with a game.cfg cross-check.
Windows	Native MakefileWin32.mak builds; the shipped ROMs are built from the Windows toolchain.
Art pipeline	Lab clustering, per-tile palettes, master sprite palettes, blue-noise dithering, halo removal, non-destructive screen fitting.
Audio	Nine ADPCM-B beds, eight FM tracks, nine SSG tracks, a recorded voice bank, pan / LFO / noise / tempo / CSM control, a working fade engine.
Desktop tools	Artbox Studio and Sound Studio, both PyQt6.

FOUND BY OBSERVATION, NOT ASSUMPTION

Engine correctness.

SPRITE TEARDOWN REWRITTEN

A full 64-word SCB1 wipe with blank tile 0xFFFF, not tile 0, which may carry art. Ghost sprites persisting across scene transitions were the visible symptom.

SPRITE SLOT PRIORITY REVERSED

HIGHER is in front. The earlier claim was the opposite; backgrounds parked at “behind” slots were drawing over characters.

BACKDROP REGISTER CORRECTED

\$401FFE. \$402000 is a mirror that silently does nothing.

FIX LAYER ROW ORIGIN CORRECTED

28 visible rows; visible row y is map row y + 2.

THE Z80 POLLING RACE FIXED

soundCommand called isZ80Ready() twice; the trailing call could deadlock the 68000 permanently: a frozen white screen with no sound.

MORE IN THE CHANGELOG

REG_PALBANK0/1 address swap, per-axis camera dead zone, START_GAME syncing to VBlank before clearing sprites, and weak fallbacks for ng_screen_table[] are also fixed. Full detail, subsystem by subsystem: CHANGELOG.md.

HONESTY OVER A CLEAN STORY

One thing tried, and removed.

An SSG envelope and three-formant phoneme speech engine was built and evaluated on hardware, then removed: a pure YM2610 SSG path cannot synthesise intelligible speech regardless of driver tricks.

THE SHIPPING ANSWER

Recorded ADPCM-A samples. speakText() is built on them: it prefers a whole-word sample, then falls back to letters, digits, and punctuation.

make GAME=<name> all make test

Self-made Neo Geo games, demos, and homebrew, close to the metal, with an optional 2D engine and a modern build loop.

REPOSITORY

github.com/ eaglesoftware777/ neogeosdk

RELEASE

v1.7.0

Source tree, six ROM sets, and the x-tools-v2 toolchain.

LICENSE

MIT

Free to modify, and ship.

For lawful homebrew development, technical research, education, preservation-oriented study, or authorised consulting work only.